PHYSICS - ELECTRIC CHARGES AND FIELDS

Chapter 1 | Comprehensive Objective Questions | Class 12 | Max Time: 120 Mins

Section A — Objective Type Questions

1. Coulomb's force is what type of force?
(A) Only Electric force (B) Only Central force (C) Both Central and Electric (D) None of these

2. Statement I: Electric lines of force do not intersect.
Statement II: The tangent at any point on a line of force represents the field direction.
(A) Both statements are correct; Statement-II explains Statement-I.
(B) Both statements are correct; Statement-II does NOT explain Statement-I.
(C) Statement-I is correct, but Statement-II is false.
(D) Statement-I is false, but Statement-II is correct.

3. The quantum of electric charge in e.s.u. unit is:
(A) 4.78×10^{-10} (B) $+16 \times 10^{-19}$
(C) 299×10^{9} (D) -16×10^{-19}

4. The nature of an electrostatic field is:
(A) Non-conservative (B) Destructive
(C) Mixed in nature (D) Conservative

5. 1 coulomb charge is equivalent to how many e.s.u.?
(A) 8.85×10^{-12} (B) 3×10^{9}
(C) 9×10^{9} (D) 1.6×10^{-19}

6. Which of the following is a vector quantity?
(A) Electric Potential (B) Electric Intensity
(C) Electric Charge (D) Electric Capacitance

7. 1 statcoulomb is equal to:
(A) 3×10^{9} C (B) $1/3 \times 10^{-9}$ C
(C) 3×10^{-9} C (D) $1/3 \times 10^{9}$ C

8. Statement I: When an object is negatively charged, its mass increases slightly.
Statement II: Charging is due to electron transfer.
(A) Both are correct; II explains I.
(B) Both are correct; II doesn't explain I.
(C) I is correct, II is incorrect.
(D) I is incorrect, II is correct.

9. If the charge on a sphere is 10μC, then the electric flux on its surface is:
(A) $36\pi \times 10^{4}\ \mathrm{Nm^2/C}$ (B) $36\pi \times 10^{-4}\ \mathrm{Nm^2/C}$
(C) $36\pi \times 10^{6}\ \mathrm{Nm^2/C}$ (D) $36\pi \times 10^{-6}\ \mathrm{Nm^2/C}$

10. The S.I. unit of electric field $\vec{E}$ is:
(A) $\mathrm{Cm^{-2}}$ (B) Vm (C) $\mathrm{JC^{-1}}$ (D) $\mathrm{NC^{-1}}$

11. An electric dipole is made of two opposite charges of $\pm 3.2 \times 10^{-19} C$ separated by $2.4 \times 10^{-10} m$. The dipole moment is:
(A) 7.68×10^{-27} Cm (B) 7.68×10^{-29} Cm
(C) 7.86×10^{-29} Cm (D) 7.86×10^{-27} Cm

12. The value of permittivity of free space (ϵ_0) is:
(A) $9 \times 10^{9}\ \mathrm{Fm^{-1}}$ (B) $8.85 \times 10^{-12}\ \mathrm{Fm^{-1}}$
(C) $1.6 \times 10^{-19}\ \mathrm{Fm^{-1}}$ (D) $8.85 \times 10^{-9}\ \mathrm{Fm^{-1}}$

13. If an electric dipole of moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$, the torque acting on it will be:
(A) $\vec{p} \cdot \vec{E}$ (B) pE (C) $\vec{p} \times \vec{E}$ (D) $\vec{p}/\vec{E}$

14. The value of electric field intensity inside a charged hollow spherical conductor is:
(A) Zero (B) $E_0\sigma$ (C) $\frac{\sigma}{E_0}$ (D) Infinite

15. The electric field near an infinitely charged thin conducting sheet is:
(A) $\frac{\sigma}{2\epsilon_0}$ (B) $\frac{\sigma}{\epsilon_0}$ (C) $\epsilon_0\sigma$ (D) $\frac{1}{2}\sigma\epsilon_0$

16. The unit of surface charge density is:
(A) Cm^{-1} (B) Cm
(C) Nm^{-1} (D) Cm^{-2}

17. If a $+q$ charge is placed inside any closed spherical surface, the total electric flux coming out is:
(A) $\frac{q}{\epsilon_0}$ (B) $q \times \epsilon_0$ (C) $\frac{\epsilon_0}{q}$ (D) $\frac{q^2}{\epsilon_0}$

18. Which of the following is NOT accelerated by an electric field?
(A) Proton (B) Neutron
(C) Alpha-particle (D) Electron

19. The work done in deflecting an electric dipole by an angle θ from the direction of uniform electric field $\vec{E}$ is:
(A) $W = pE(1 - \sin\theta)$ (B) $W = pE(1 - \cos\theta)$
(C) $W = pE\tan\theta$ (D) $W = pE\sec\theta$

20. The relative permittivity (ϵ_r) of a medium is:
(A) $\epsilon + \epsilon_0$ (B) $\epsilon \times \epsilon_0$ (C) $\frac{\epsilon}{\epsilon_0}$ (D) $\epsilon - \epsilon_0$

21. The value of $\frac{1}{4\pi\epsilon_0}$ is:
(A) $9 \times 10^{-9}\ \mathrm{Nm^2C^{-2}}$ (B) $9 \times 10^{9}\ \mathrm{Nm^2C^{-2}}$
(C) $9 \times 10^{12}\ \mathrm{Nm^2C^{-2}}$ (D) $9 \times 10^{-12}\ \mathrm{Nm^2C^{-2}}$

22. The correct relation between electric field $\vec{E}$ and electric force $\vec{F}$ on a charge q is:
(A) $\vec{E} = \frac{q}{\vec{F}}$ (B) $\vec{E} = q\vec{F}$ (C) $\vec{E} = \frac{\vec{F}}{q}$ (D) $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{\vec{F}}$

23. The S.I. unit of electric dipole moment is:
(A) cm^2 (B) cm^{-1} (C) cm^{-2} (D) Cm

24. Which of the following values of n is not possible in the relation $Q = ne$?
(A) 4.2 (B) 8 (C) 4 (D) 100

25. The correct value of the elementary charge on an electron is:
(A) $1.602 \times 10^{-16} C$ (B) $1.602 \times 10^{-19} C$
(C) $1.602 \times 10^{-18} C$ (D) $1.602 \times 10^{-17} C$

26. The correct vector form of Coulomb's Law is:
(A) $\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{|\vec{r}|^3}$ (B) $\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{|\vec{r}|^2}\vec{r}$
(C) $\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{|\vec{r}|^3}\vec{r}$ (D) $\vec{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\vec{r}$

27. The total electric flux coming out from a stationary unit positive charge in free space is:
(A) ϵ_0 (B) $4\pi\epsilon_0$ (C) $(4\pi\epsilon_0)^{-1}$ (D) ϵ_0^{-1}

28. Quantization of charge implies that:
(A) A fraction of charge on an electron is not possible
(B) A charge cannot be destroyed
(C) Charges exist only on particles
(D) Charge values can be infinite

29. The ratio of the electric force and gravitational force acting between an electron and a proton is in the order of:
(A) 10^{42} (B) 10^{36} (C) 10^{39} (D) 1

30. A charge 'q' is placed at the center of the line joining two equal charges 'Q'. For the system to be in equilibrium, 'q' must be:
(A) $-\frac{Q}{2}$ (B) $-4Q$ (C) $+\frac{Q}{4}$ (D) $-\frac{Q}{4}$

31. For an electric dipole to be in stable equilibrium in a uniform electric field, the angle between $\vec{P}$ and $\vec{E}$ should be:
(A) 0° (B) 90° (C) 180° (D) 45°

32. The number of electrons present in 8 Coulomb of negative charge is:
(A) 12.8×10^{19} (B) 5×10^{19}
(C) 2.5×10^{19} (D) 1.6×10^{19}

33. The electric field at the center of a solid charged conducting sphere is:
(A) Maximum (B) Zero (C) Minimum (D) Infinite

34. Two equal and opposite charges of 5 C are placed 5.0 cm apart. The electric dipole moment of the system is:
(A) 5×10^{-2} Cm (B) 1 Cm
(C) 25×10^{-2} Cm (D) Zero

35. Equal charges are placed at three corners of a square. If force between adjacent charges is F_{12} and diagonal is F_{13}, then $\frac{F_{12}}{F_{13}}$ is:
(A) 2 (B) $1/2$ (C) $\sqrt{2}$ (D) $1/\sqrt{2}$

36. When 10^{14} electrons are removed from a neutral metallic sphere, the charge on the sphere becomes:
(A) $-16\mu C$ (B) $32\mu C$ (C) $-32\mu C$ (D) $+16\mu C$

37. The dimensional formula of electric flux is:
(A) $[ML^3T^{-3}A^{-1}]$ (B) $[ML^2T^{-3}A^{-1}]$ (C) $[ML^2T^{-2}A^{-2}]$ (D) $[ML^3T^{-2}A^{-1}]$

38. The charging of a body is primarily due to the transfer of:
(A) Protons (B) Electrons
(C) Neutrons (D) Positrons

39. If the distance between two point charges is doubled, the electrostatic force acting between them becomes:
(A) 2 times (B) 4 times
(C) 1/4 times (D) 1/2 times

40. The electric field near a uniformly charged infinite plane non-conducting sheet is:
(A) σ/ϵ_0 (B) $2\sigma/\epsilon_0$ (C) $\sigma/2\epsilon_0$ (D) Zero

41. Gauss's law is applicable for determining electric flux over:
(A) Open surfaces
(B) Point charges only
(C) Spherical surfaces only
(D) Closed surfaces

42. When charge is given to a solid non-conducting sphere, it:
(A) Spreads entirely on the surface
(B) Stays exactly where it was placed
(C) Resides only deep inside
(D) Resides uniformly inside and on the surface

43. A stationary electric charge produces:
(A) Only a magnetic field (B) Only an electric field
(C) Both fields (D) Neither fields

44. The ratio of the magnitude of electric field due to a short dipole at equal distances in the axial to the equatorial positions is:
(A) 1:2 (B) 1:1 (C) 2:1 (D) 4:1

45. The direction of the electric field on the equatorial line of an electric dipole with respect to its dipole moment vector is:
(A) Same (B) Opposite
(C) Perpendicular (D) At an angle of 45°

46. 1 Volt/meter is equivalent to which of the following units?
(A) 1 J/C (B) 1 V/C (C) 1 N/C (D) 1 C/N

47. If a dielectric medium of constant K is placed between two charges, the electrostatic force between them compared to vacuum:
(A) Becomes K times (B) Becomes $1/K$ times
(C) Becomes K^2 times (D) Remains unchanged

48. A charge q is placed exactly at the center of a cube. The electric flux coming out through any ONE face of the cube will be:
(A) q/ϵ_0 (B) $q/2\epsilon_0$ (C) $q/6\epsilon_0$ (D) $q/4\epsilon_0$

49. A proton and an electron are placed in a uniform electric field. Which of the following statements is true regarding the forces acting on them?
(A) Their directions will be the same
(B) Their magnitudes will be exactly equal
(C) Their accelerations will be exactly equal
(D) Their magnitudes will be completely different

50. The electric field at a point due to an infinitely long straight uniformly charged wire is proportional to:
(A) $1/r^2$ (B) r (C) $1/r$ (D) $1/r^3$

51. An electric dipole placed in a non-uniform electric field will generally experience:
(A) Only a net force
(B) Both a net force and a torque
(C) Only a torque
(D) Neither a force nor a torque

52. 1 micro-coulomb ($1\mu C$) is equal to:
(A) $10^{-6}C$ (B) $10^{-3}C$ (C) $10^{-9}C$ (D) $10^{-12}C$

53. Two like charges naturally do what to each other?
(A) Attract (B) Apply no force (C) Repel (D) Destroy

54. When a glass rod is rubbed with silk, the glass rod loses electrons and becomes:
(A) Negatively charged (B) Positively charged
(C) Neutral (D) Both pos and neg

55. Which of the following charge values is not possible on a body?
(A) $3.2 \times 10^{-19}C$ (B) $6.4 \times 10^{-19}C$
(C) $1.6 \times 10^{-19}C$ (D) $2.4 \times 10^{-19}C$

56. At what angle do two electric field lines intersect each other?
(A) 90° (B) 45°
(C) They never intersect (D) 30°

57. When two charged conductors are connected by a wire, charge will not flow between them if:
(A) Their capacitances are equal
(B) Their electrostatic potentials are equal
(C) Their total charges are equal
(D) Their physical sizes are equal

58. Electric field lines on the surface of a charged conductor are always:
(A) Perpendicular to the surface
(B) Parallel to the surface
(C) At 45° to the surface
(D) In any random direction

59. The electric field (E) due to a short electric dipole at a distance r is proportional to:
(A) $1/r$ (B) $1/r^2$ (C) $1/r^3$ (D) r^2

60. When an electron is brought closer to another electron, the electrostatic potential energy:
(A) Decreases (B) Becomes zero
(C) Remains unchanged (D) Increases

61. The total electric flux over a closed surface in vacuum is what times the total net charge enclosed within it?
(A) $1/\epsilon_0$ times (B) ϵ_0 times
(C) $4\pi\epsilon_0$ times (D) $1/4\pi\epsilon_0$ times

62. The conventional direction of the electric dipole moment ($\vec{p}$) is:
(A) From positive charge to negative charge
(B) Perpendicular to the electric field axis
(C) Not fixed in any frame
(D) From negative charge to positive charge

63. The dimensional formula of electric charge is:
(A) $[AT]$ (B) $[AT^{-1}]$ (C) $[A^{-1}T]$ (D) $[AT^2]$

64. The absolute value of the elementary charge (e) is:
(A) $9.1 \times 10^{-31}C$ (B) $1.6 \times 10^{-19}C$
(C) $1.6 \times 10^{-27}C$ (D) $8.85 \times 10^{-12}C$

65. The intensity of the electric field just outside the surface of a charged conductor is:
(A) Zero
(B) Tangential to the surface
(C) Perpendicular to the surface
(D) At an angle of 45°

66. If a dielectric medium is introduced between two point charges, the electrostatic force:
(A) Increases (B) Decreases
(C) Remains completely unchanged (D) Automatically becomes zero

67. The potential energy of an electric dipole in an external uniform electric field is minimum when the angle between $\vec{p}$ and $\vec{E}$ is:
(A) 90° (B) 180° (C) 0° (D) 270°

68. How does the electric field E due to an infinite non-conducting plane sheet of charge depend on the distance r?
(A) Directly proportional to r
(B) Inversely proportional to r^2
(C) Inversely proportional to r
(D) Independent of distance r

69. When a soap bubble is given a negative charge, its radius:
(A) Decreases (B) Remains unchanged
(C) Becomes zero (D) Increases

70. The value of dielectric constant (K) for metallic conductors is:
(A) 0 (B) Infinity (∞) (C) 1 (D) -1

71. The number of extra electrons on a body required to give it a net charge of -1 Coulomb is:
(A) 1.6×10^{-19} (B) 6.25×10^{19}
(C) 6.25×10^{18} (D) 1.6×10^{19}

72. The torque on a dipole placed in a uniform electric field is maximum when the angle is:
(A) 0° (B) 90° (C) 45° (D) 180°

73. In Coulomb's law, the value of the constant $k = 1/(4\pi\epsilon_0)$ depends on:
(A) Only the nature of the intervening medium
(B) Only the system of units adopted
(C) Neither the medium nor the units
(D) Both the nature of the medium and the system of units

74. The principle of conservation of charge implies that:
(A) The total charge of the universe is always zero
(B) The total net charge of an isolated system remains constant
(C) Electric charge can only be destroyed
(D) Electric charge can only be created

75. The work done in moving a test charge from one point to another on an equipotential surface is:
(A) Positive (B) Negative (C) Zero (D) Infinite

76. When a charged particle is released from rest in a uniform electric field, its kinetic energy:
(A) Decreases continually (B) Remains constant
(C) Becomes zero (D) Increases continually

77. Gauss's law in electrostatics is valid only if the electrostatic force is inversely proportional to which power of distance r?
(A) r^3 (B) r^2 (C) r (D) r^{-2}

78. The electric field inside a hollow conducting sphere is exactly zero. This phenomenon is commonly known as:
(A) Electrostatic shielding
(B) Dielectric Polarization
(C) Dielectric breakdown
(D) Electromagnetic Induction

79. If the product $q_1q_2 > 0$, the nature of the electrostatic force between the two charges will be:
(A) Attractive (B) Zero
(C) Repulsive (D) Cannot be determined

80. If the product $q_1q_2 < 0$, the nature of the electrostatic force between the two charges will be:
(A) Repulsive (B) Attractive
(C) Zero (D) Cannot be determined

81. Electric field lines start from which charge and end on which charge?
(A) From negative to positive
(B) From infinity to positive
(C) From positive to negative
(D) They form continuous closed loops

82. Two electric field lines can never intersect each other primarily because:
(A) Field lines are just imaginary paths
(B) At intersection, there would be two directions of electric field
(C) They constantly repel each other
(D) They are always parallel to each other

83. The ratio of electric potential (V) to electric field (E) at a distance r from a point charge q is:
(A) $1/r$ (B) r^2 (C) r (D) $1/r^2$

84. Electron volt (eV) is a practical unit of:
(A) Electric Charge (B) Electric Potential
(C) Electric Current (D) Energy

85. An electron placed in a uniform electric field will experience an electrostatic force in which direction?
(A) Perpendicular to the field
(B) In the direction of the field
(C) Opposite to the direction of the field
(D) At an angle of 45°

86. The net electrostatic force on an electric dipole placed in a uniform electric field is always:
(A) Zero (B) pE (C) $2qE$ (D) qE

87. The amount of charge on any body is always an integer multiple of which fundamental value?
(A) $3.2 \times 10^{-19}C$ (B) $1.6 \times 10^{-19}C$
(C) $9.1 \times 10^{-31}C$ (D) 1 Coulomb

88. The direction of the electrostatic force between two point charges is always:
(A) In any random orientation
(B) Perpendicular to the line joining them
(C) Along the line joining them
(D) Tangential to their surfaces

89. The dielectric strength of normal dry air is approximately:
(A) 3×10^6 Vm^{-1} (B) 3×10^8 Vm^{-1}
(C) 1×10^6 Vm^{-1} (D) 9×10^9 Vm^{-1}

90. Electric flux is which type of physical quantity?
(A) Vector (B) Scalar (C) Tensor (D) None

91. If the total electric flux coming out of a closed surface is zero, it definitively implies that:
(A) There is no net charge enclosed by the surface
(B) There is only positive charge inside
(C) There is only negative charge inside
(D) There is absolutely no charge outside the surface

92. The net electric charge on an alpha particle (α-particle) is:
(A) $-1.6 \times 10^{-19}C$ (B) $+1.6 \times 10^{-19}C$
(C) $+3.2 \times 10^{-19}C$ (D) Zero

93. The total electric charge on a neutron is:
(A) $+1.6 \times 10^{-19}C$ (B) Zero
(C) $-1.6 \times 10^{-19}C$ (D) $+3.2 \times 10^{-19}C$

94. A charged particle of mass m and charge q is placed in a uniform electric field E. Its acceleration will be:
(A) mE/q (B) q/mE (C) qE/m (D) m/qE

95. The total electric flux passing through a closed spherical surface enclosing an electric dipole is:
(A) q/ϵ_0 (B) $2q/\epsilon_0$ (C) Infinite (D) Zero

96. The surface charge density at the sharp points of an irregularly shaped charged conductor is:
(A) Maximum (B) Minimum (C) Zero (D) Uniform

97. The absolute value of 1 nanocoulomb (1 nC) is:
(A) $10^{-6}C$ (B) $10^{-9}C$ (C) $10^{-12}C$ (D) $10^{-3}C$

98. If 3 separate electric dipoles are placed inside a closed surface, the total electric flux coming out of the surface will be:
(A) $3q/\epsilon_0$ (B) $6q/\epsilon_0$ (C) Zero (D) q/ϵ_0

99. The electric field exactly at the center of a uniformly charged thin spherical shell is:
(A) σ/ϵ_0 (B) Infinite (C) $q/4\pi\epsilon_0R^2$ (D) Zero

100. When a proton is brought closer to another proton, the electrostatic potential energy of the system:
(A) Becomes exactly zero (B) Decreases
(C) Increases (D) Remains unchanged